SUVAT – Freefall Problems

For each question, draw a simple diagram, choose which direction you will take as positive, and list the SUVAT variables you know before starting. Then choose an appropriate equation and solve the problem.

Take the acceleration due to gravity as

$$g = 9.8\,\mathrm{m\,s^{-2}}$$

and ignore air resistance.

1. A ball is dropped from rest. Calculate its velocity after 3.0 s.

2. A ball is thrown vertically upwards with an initial velocity of $19.6\,\mathrm{m\,s^{-1}}$. Calculate the time taken for it to reach its maximum height.

3. A ball is thrown vertically upwards with an initial velocity of $21\,\mathrm{m\,s^{-1}}$. Calculate the maximum height gained by the ball.

4. A stone is dropped from rest from a height of 40 m. Calculate its speed immediately before it reaches the ground.

5. A ball is thrown vertically downwards with an initial velocity of $5.0\,\mathrm{m\,s^{-1}}$. Calculate the distance it falls in 2.0 s.

6. A ball is thrown vertically upwards with an initial velocity of $15\,\mathrm{m\,s^{-1}}$. Calculate its displacement after 1.2 s.

7. A stone is dropped from rest and falls a vertical distance of 19.6 m. Calculate the time taken to fall this distance.

8. A falling stone has a downward velocity of $6.0\,\mathrm{m\,s^{-1}}$. Two seconds later, its downward velocity is $25.6\,\mathrm{m\,s^{-1}}$. Calculate the distance travelled by the stone during these 2.0 s.

9. A ball is thrown vertically upwards with an initial velocity of $24\,\mathrm{m\,s^{-1}}$. Calculate its velocity 3.0 s after being released.

10. While travelling upwards, a ball slows from $18\,\mathrm{m\,s^{-1}}$ to $8.2\,\mathrm{m\,s^{-1}}$ in 1.0 s. Calculate the vertical distance travelled during this time.

Answers

1. $29.4\,\mathrm{m\,s^{-1}}$ downwards
2. $2.0\,\mathrm{s}$
3. $22.5\,\mathrm{m}$
4. $28.0\,\mathrm{m\,s^{-1}}$
5. $29.6\,\mathrm{m}$
6. $10.9\,\mathrm{m}$ upwards
7. $2.0\,\mathrm{s}$
8. $31.6\,\mathrm{m}$
9. $5.4\,\mathrm{m\,s^{-1}}$ downwards
10. $13.1\,\mathrm{m}$